Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
3	(a)	(i)		same concentration / strength (1) same temperature (1) neutral answer - reference to volume or amount of acid	2			2		2
		(ii)		**B** **A** **D** **C** order correct (1) award (1) for any of following the more reactive (the metal) the more bubbles / fizzing the less reactive (the metal) the less bubbles / fizzing the most reactive (metal) has the most bubbles / fizzing the least reactive (metal) has the least bubbles / fizzing accept reference to **B** having the <u>most</u> bubbles or **C** having <u>no bubbles</u> / the <u>least</u> bubbles if correct order given		2		2		2

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)	(i)		thermometer is not in the acid / solution / in the air (1) award (1) for any of following will not measure the <u>temperature</u> of the reaction mixture / acid / solution will measure the <u>temperature</u> of the air will not get a <u>temperature</u> rise / change <u>temperature</u> will be lower than expected <u>temperature</u> recording will be incorrect neutral answers unreliable / inaccurate results thermometer gives wrong reading different temperature			2	2		2
		(ii)	I	metal **D** test **2** **both** needed accept correct answer circled in table		1		1		1
			II	23 accept 22 / 24 accept correct answer written in table		1		1	1	1
			III	accept any value in the range 14.1 to 21.9 accept any range that falls within this range e.g. 15-21			1	1	1	

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(c)	(i)		$Mg + \boxed{2} HCl \longrightarrow MgCl_2 + H_2$ award (1) for formula award (1) for balancing only if formula is correct		2		2		
		(ii)		magnesium nitrate accept $Mg(NO_3)_2$		1		1		
				Question 3 total	**2**	**7**	**3**	**12**	**2**	**8**